

Association Rule Learning on the Groceries Dataset

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Assignment

Question 1: Understand the Data

- a. Number of transactions: 9,835
- b. Number of unique items: 169
- c. The 10 most frequently purchased items:

Rank	Item	Count
1	whole milk	2513
2	other vegetables	1903
3	rolls/buns	1809
4	soda	1715
5	yogurt	1372
6	bottled water	1087
7	root vegetables	1072
8	tropical fruit	1032
9	shopping bags	969
10	sausage	924

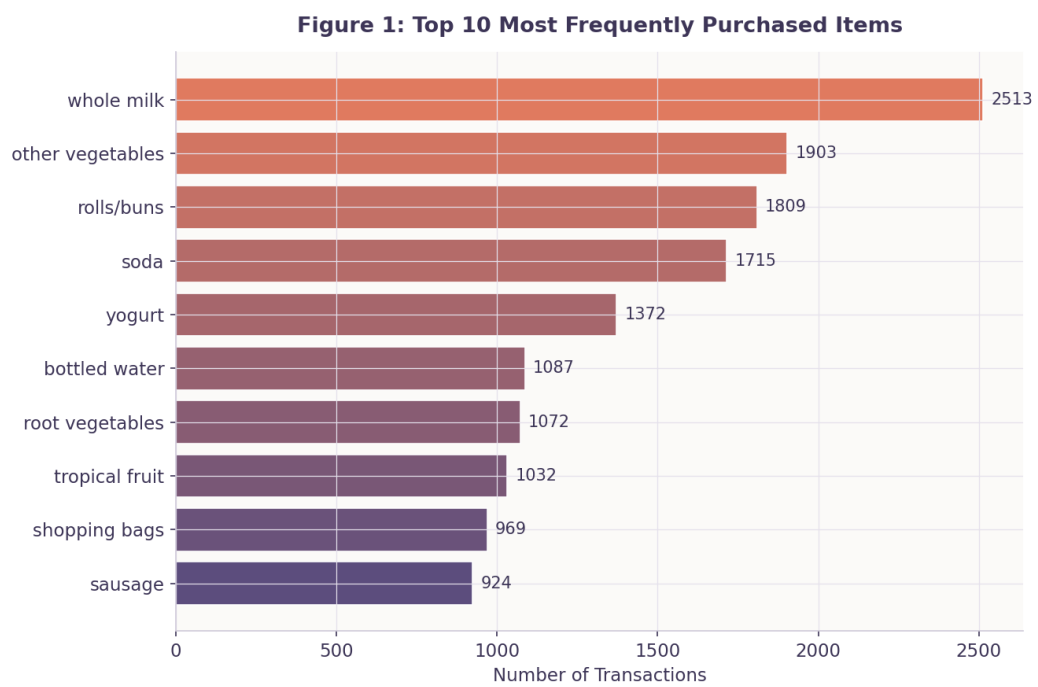


Figure 1: Top 10 most frequently purchased items.

Question 2: Frequent Itemsets

Using the Apriori algorithm with a minimum support threshold of 0.01, the dataset was searched for itemsets of two or more products that co-occur frequently. This produced 213 frequent pairs and 32 frequent triples.

a & b. At least 10 frequent itemsets (2+ products) with support values:

Rank	Itemset	Support
1	other vegetables, whole milk	0.0748
2	rolls/buns, whole milk	0.0566
3	yogurt, whole milk	0.0560
4	root vegetables, whole milk	0.0489
5	root vegetables, other vegetables	0.0474
6	yogurt, other vegetables	0.0434
7	rolls/buns, other vegetables	0.0426
8	tropical fruit, whole milk	0.0423
9	soda, whole milk	0.0401
10	soda, rolls/buns	0.0383

c. Itemset with the highest support: {other vegetables, whole milk}, support = 0.0748.

d. What the support value tells us about purchasing behavior: Support measures how often an itemset appears across all transactions — essentially, how common or popular that combination is. The highest-support itemsets here are dominated by whole milk, which is also individually the most frequently purchased item overall. This reflects everyday, routine grocery shopping rather than a special or surprising relationship between products — customers buy these staples on most trips regardless of what else is in the basket. Support alone only reflects popularity; it does not indicate whether two items are meaningfully associated, which is why confidence and lift (Question 3 onward) are needed to assess the true strength of a relationship.

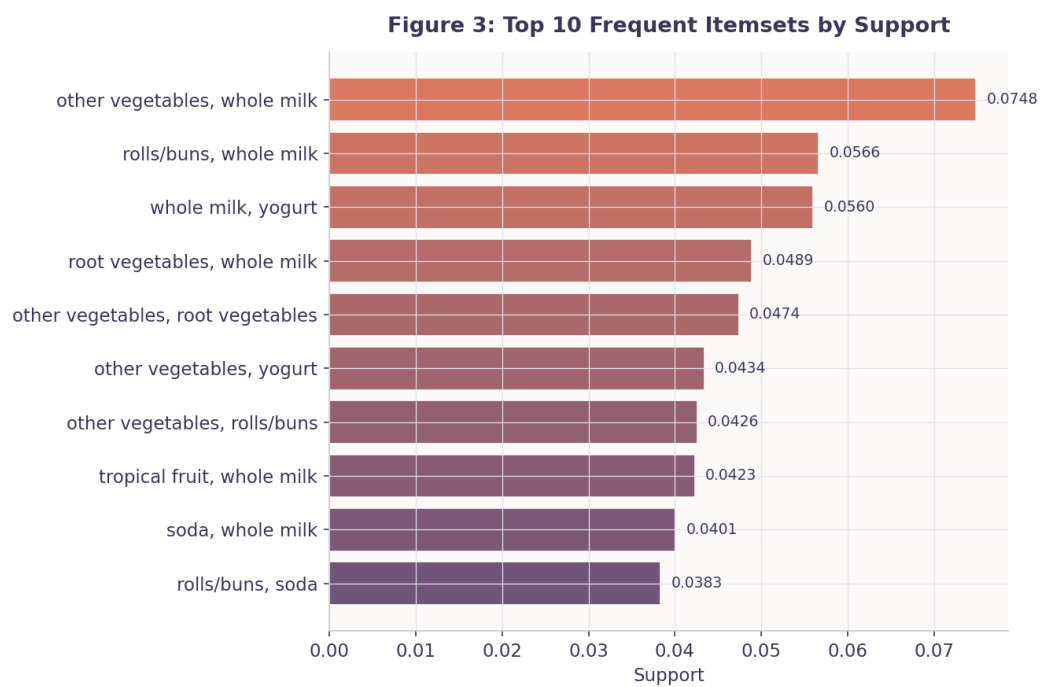


Figure 2: Top 10 frequent itemsets by support.

Question 3: Association Rules

Association rules were generated from the frequent itemsets using a minimum confidence threshold of 30%. This produced 125 rules in total; the top 10 (by confidence) are shown below.

Antecedent	Consequent	Support	Confidence	Lift
citrus fruit, root vegetables	other vegetables	0.0104	0.586	3.030
butter	whole milk	0.0276	0.497	1.946
curd	whole milk	0.0261	0.490	1.918
root vegetables, whole milk	other vegetables	0.0489	0.475	2.451
root vegetables, other vegetables	whole milk	0.0489	0.489	1.913
domestic eggs	whole milk	0.0300	0.463	1.812
whipped/sour cream	whole milk	0.0322	0.450	1.760
tropical fruit, other vegetables	whole milk	0.0170	0.446	1.745
pip fruit	whole milk	0.0301	0.418	1.635
citrus fruit, other vegetables	root vegetables	0.0104	0.359	3.295

a. At least 10 association rules generated: 125 total rules met the 30% confidence threshold; the top 10 (by confidence) are shown above.

b. For each rule — antecedent, consequent, support, confidence, lift: all reported in the table above.

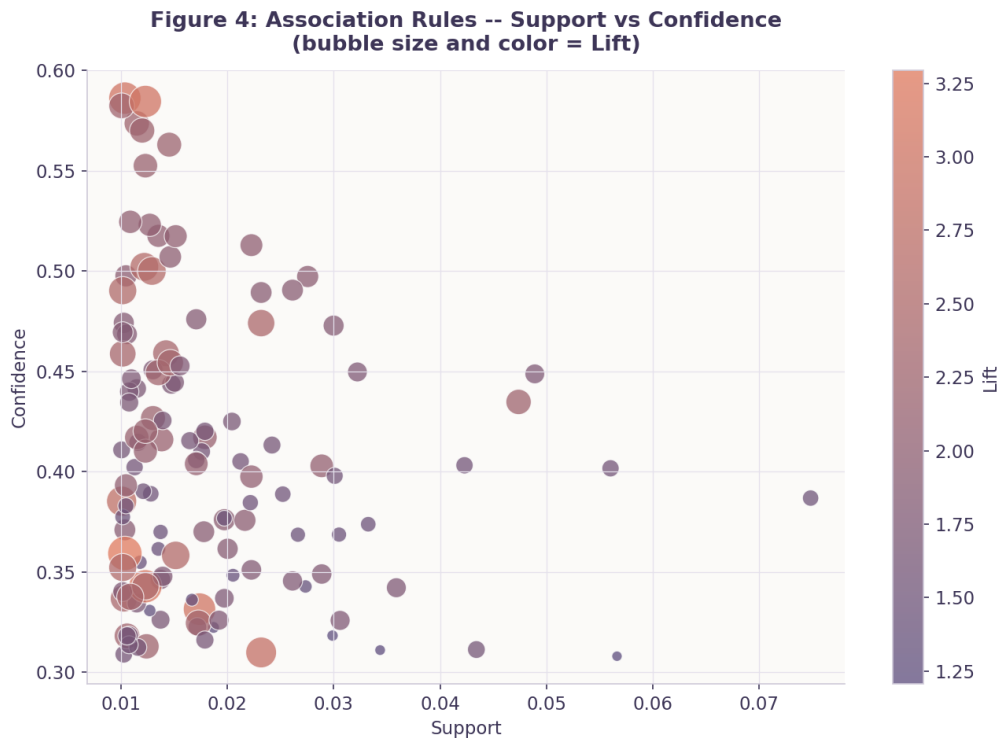


Figure 3: Association rules — support vs. confidence (bubble size/color = lift).

Question 4: Understanding the Measures

Rule 1: {beef} $\rightarrow$ {root vegetables}

Support = 0.0174, Confidence = 0.331, Lift = 3.04

- **Support:** Beef and root vegetables were bought together in 1.74% of all transactions.
- **Confidence:** Of customers who bought beef, 33.1% also bought root vegetables in the same trip.
- **Lift:** Customers who bought beef were about 3 times more likely to also buy root vegetables than a random customer — a strong, non-coincidental association (likely for stews/roasts).

Rule 2: {butter} $\rightarrow$ {whole milk}

Support = 0.0276, Confidence = 0.497, Lift = 1.95

- **Support:** Butter and whole milk appeared together in 2.76% of all transactions.
- **Confidence:** Of customers who bought butter, 49.7% also bought whole milk.
- **Lift:** Buying butter makes a customer almost twice as likely to also buy whole milk compared to chance — a solid, meaningful association.

Rule 3: {bottled water} $\rightarrow$ {whole milk}

Support = 0.0344, Confidence = 0.311, Lift = 1.22

- **Support:** Bottled water and whole milk appeared together in 3.44% of all transactions.
- **Confidence:** Of customers who bought bottled water, 31.1% also bought whole milk.
- **Lift:** A lift of only 1.22 (close to 1) means bottled water barely increases the likelihood of buying whole milk beyond chance — a weak association, since whole milk is bought so often by everyone anyway.

Question 5: Find Interesting Rules

a. **Highest confidence:** {citrus fruit, root vegetables} $\rightarrow$ {other vegetables}, confidence = 0.586

b. **Highest lift:** {citrus fruit, other vegetables} $\rightarrow$ {root vegetables}, lift = 3.295

c. **Highest support:** {other vegetables} $\rightarrow$ {whole milk}, support = 0.0748

d. **Are these the same rule? Explain:** No — all three are different rules, and this is expected rather than a coincidence, because each metric captures something different. Support rewards item-sets that are simply popular overall: {other vegetables} and {whole milk} win here because they are individually the two most commonly bought items in the dataset, even though the relationship between them is not especially strong (lift only 1.51). Confidence rewards rules where the consequent reliably follows the antecedent, regardless of overall popularity. Lift rewards the strongest genuine association relative to chance, independent of how common the items are. A rule can therefore be common (high support) without being a strong relationship, and a rare combination (low support) can still represent the strongest true association in the dataset — which is exactly why all three metrics should be considered together rather than relying on any single one.

Question 6: Interpret a Rule

Selected rule: {citrus fruit, other vegetables} $\rightarrow$ {root vegetables} (Support = 0.0104, Confidence = 0.359, Lift = 3.295)

In simple language: Customers who purchase citrus fruit and other vegetables together also tend to purchase root vegetables. Of all customers who bought citrus fruit and other vegetables in the same trip, about 36% also bought root vegetables — roughly 3.3 times more often than would be expected if these purchases were unrelated.

Strong or weak association, and why: This is a strong association. The lift of 3.295 is well above 1, meaning the relationship is unlikely to be coincidental — customers buying citrus fruit and other vegetables together are genuinely more inclined to also buy root vegetables, likely because all three are fresh produce commonly used together (e.g. for a stew, salad, or juicing). The one caveat is that support is low (1.04%), meaning this exact three-item combination is relatively rare overall, so while the relationship is strong when it occurs, it is a niche rather than everyday pattern.

Question 7: Business Application

a. Two products to place close to each other: Beef and root vegetables. This pair has a high lift (3.04, from Question 4) — customers who buy beef are about 3 times more likely to also buy root vegetables than chance predicts. Placing root vegetables near the meat counter would make it easier for beef shoppers to notice and add them, likely increasing basket size.

b. One possible product bundle: A “vegetable stew” bundle of citrus fruit, other vegetables, and root vegetables, based on the highest-lift rule found in Question 5/6 (lift = 3.295). A discounted “Fresh Stew Starter Pack” combining these items could appeal to customers who already tend to buy this exact combination.

c. Example of using association rules for recommendations: The rule {butter} $\rightarrow$ {whole milk} (confidence 0.497, lift 1.95) could power a “customers who bought this also bought...” recommendation at checkout or in a mobile app. When a customer adds butter to their basket, the system could automatically suggest whole milk — since the rule has both decent confidence and meaningful lift, the suggestion is likely to feel relevant rather than random.

Question 8: Understanding Association

a. Does an association rule prove causation? No. Association rules only show that two products tend to appear together in the same transactions more (or less) often than chance would predict — they say nothing about why. A rule like {butter} $\rightarrow$ {whole milk} does not mean buying butter causes someone to buy whole milk; both may simply be common staples bought on the same routine trip. Association rule mining is a correlational technique, not a causal one; establishing causation would require controlled experiments or further causal analysis.

b. Why might a rule have high confidence but low lift? This happens when the consequent item is extremely popular on its own, regardless of the antecedent. Confidence only measures how often the consequent follows the antecedent, without accounting for how often the consequent would appear anyway. If the consequent (e.g. whole milk) is bought in a large share of all transactions, almost any antecedent will show reasonably high confidence toward it simply because the consequent is everywhere. Lift corrects for this by comparing confidence against the consequent’s baseline

popularity, so a rule can look “reliable” (high confidence) while actually being close to random chance (lift near 1). This was seen with $\{\text{bottled water}\} \rightarrow \{\text{whole milk}\}$: confidence 0.311 sounds decent, but lift was only 1.22.

c. What does a lift value greater than 1 indicate? A lift greater than 1 indicates a positive association — the two items are purchased together more often than would be expected if they were bought independently. The further above 1, the stronger the association (e.g. lift = 3 means the pair co-occurs about 3 times more often than chance). A lift of exactly 1 means no relationship (statistical independence), and a lift below 1 indicates the items are bought together less often than expected — a negative association.

Limitations

1. The dataset does not contain customer identifiers, so customer-level purchasing behavior cannot be studied.
2. There are no timestamps, so seasonal or time-of-day purchasing patterns cannot be analyzed.
3. Prices, quantities, and profit margins are not available, so the financial value of a rule cannot be assessed.
4. Association rules describe correlation, not causation.
5. Results depend on the chosen support and confidence thresholds.
6. Some rules have high lift but low support, meaning they are statistically strong but numerically rare.

Summary

This analysis used the Apriori algorithm and association rule mining on the Groceries dataset (9,835 transactions, 169 unique items) to identify frequently co-purchased products and the strength of their relationships. Whole milk and other vegetables dominate in raw frequency and support, but the most genuinely interesting relationships — as measured by lift — involve fresh produce combinations such as citrus fruit, other vegetables, and root vegetables. These findings suggest practical applications for product placement, bundling, and recommendation systems in a supermarket setting.

References

1. Agrawal, R., & Srikant, R. (1994). *Fast Algorithms for Mining Association Rules*. Proceedings of the 20th VLDB Conference.
2. Raschka, S. (2018). *MLxtend: Providing machine learning and data science utilities and extensions to Python’s scientific computing stack*. Journal of Open Source Software, 3(24), 638.